CLASS 11.03.2026

Reversible transition (2 concentrations)

$$A \underset{\lambda_b}{\overset{\lambda_a}{\rightleftarrows}} B \quad , \lambda_a, \lambda_b > 0$$

If we consider a perturbation of the initial condition of the equilibrium $(\bar{a}, \bar{b})$, i.e.

$$a(0) = a_0 + a_1 > 0, \qquad b(0) = b_0 + b_1 > 0$$

then (a, b) evolves until it reaches another equ $(\tilde{a}, \tilde{b})$ s.t.

$$\begin{cases} \lambda_a \tilde{a} = \lambda_b \tilde{b} \\ \tilde{a} + \tilde{b} = a_0 + a_1 + b_0 + b_1 \end{cases}$$

$$\begin{cases} \tilde{a} = \dfrac{\lambda_b}{\lambda_a}\,\tilde{b} \\ \tilde{b}\left(1+\dfrac{\lambda_b}{\lambda_a}\right) = a_0+a_1+b_0+b_1 \end{cases} \Rightarrow \begin{cases} \tilde{a} = \dfrac{\lambda_b}{\lambda_a+\lambda_b}\left(a_0+b_0+a_1+b_1\right) \\ \tilde{b} = \dfrac{\lambda_a}{\lambda_a+\lambda_b}\left(a_0+b_0+a_1+b_1\right) \end{cases}$$

Le Chatelier principle

If a chemical system at the equilibrium suffers a change of concentration / local pressure / tempered, then the system evolves to reach a new equilibrium.

Reversible transition (m concentrations)

$$X_i \underset{\lambda_j^i}{\overset{\lambda_i^j}{\rightleftharpoons}} X_j \quad , \quad i,j \in \{1,\dots,m\} \quad i \neq j$$

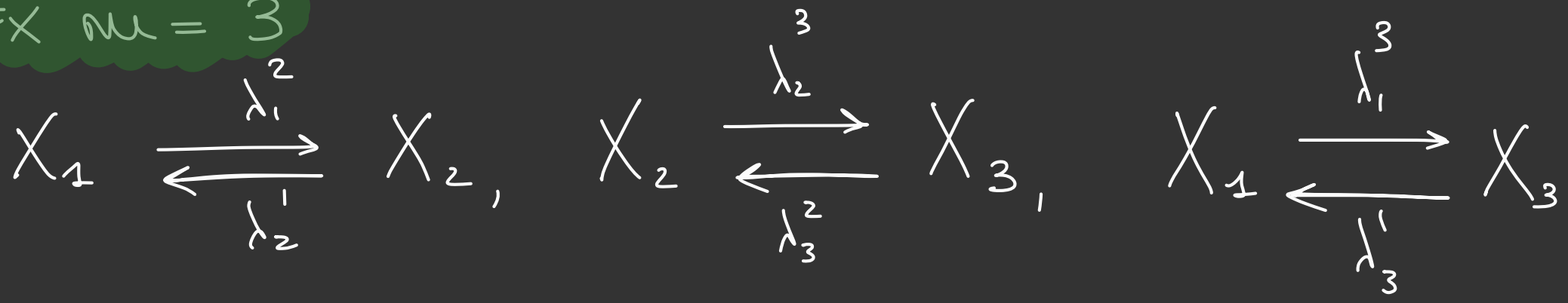
$$\lambda_{ij} : \begin{cases} > 0, & i \neq j \\ 0, & i = j \end{cases} \quad \rightsquigarrow \text{ (no trasformation)}$$

$$x := (x_1, \dots, x_m)$$

$$\rightsquigarrow \quad \frac{d}{dt} x = - \Lambda x$$

with $$\Lambda := \begin{bmatrix} -\sum_{i \neq 1} \lambda_1^i & \lambda_2^1 & \lambda_3^1 & \dots & \lambda_m^1 \\ \lambda_1^2 & -\sum_{i \neq 2} \lambda_2^i & \lambda_3^2 & \dots & \lambda_m^2 \\ \lambda_1^3 & \ddots & \ddots & \ddots & \vdots \\ \vdots & & & & \lambda_{m-1}^{m-1} \\ \lambda_1^m & \lambda_2^m & \dots & \lambda_{m-1}^m & -\sum_{i \neq m} \lambda_m^i \end{bmatrix}$$

Ex $n = 3$



$$\Rightarrow \quad \dot{x} = \begin{bmatrix} -\lambda_1^2 - \lambda_1^3 & \lambda_2^1 & \lambda_3^1 \\ \lambda_1^2 & -\lambda_2^1 - \lambda_2^3 & \lambda_3^2 \\ \lambda_1^3 & \lambda_2^3 & -\lambda_3^1 - \lambda_3^2 \end{bmatrix} x$$

$$(a_i^j) = A =: \quad i,j = 1,\dots m$$

* symmetry: iff each reaction is symmetric $\left(\lambda_i^j = \lambda_j^i \;\; i \neq j\right)$
* $\forall j \quad \sum_i a_i^j = 0$ sum of elements of each column $= 0$

Therefore

$$\frac{d}{dt}\left(\sum_i x_i\right) = \sum_i \frac{dx_i}{dt}$$

$$= \sum_i^m \left(- \sum_{j\neq i}^m \lambda_i^j x_i \right) + \sum_i^m \sum_{j\neq i} \lambda_j^i x_j$$

$$= - \sum_{\substack{i,j \\ i\neq j}} \lambda_i^j x_i + \sum_{\substack{i,j \\ i\neq j}} \lambda_j^i x_j = 0$$

$\Rightarrow$ $\sum$ concentrations are conserved in time.

DEF: Dynamical system

it is a mathematical model that describes an evolution process.

It is caracterised by

* set of states S : state of elements that represent the state of the system.

also called phase space

ex $S = \{x, \dot{x}, \ddot{x}, \dots, x^{n}\} \subseteq \mathbb{R}^{n+1}$

** time t : a parameter $\in \mathcal{T} \subseteq \mathbb{R}$ (independent variable)

*** evolution law : describes the evolution of the states of the system.

Rmk ** can be a discrete or continuous variable

Namely, we are going to study CLASSICAL DYNAMICAL SYSTEM

DEF:

1) S is a differentiable manifold with finite dimension

Ex $S = M \subset C^{\infty}$ or $S = D \subseteq \mathbb{R}^{n}$

and its structure is preserved by the evolution law

2) Determinism and evolution reversibility: (the system is globally well posed) $\mathcal{T} = \mathbb{R}$ or $\mathcal{T} = \mathbb{Z}$

3) the evolution law does not change in time

⚠ why do I assume S as a differentiable manifold? → ex. open domain or domain that admit a system of local coordinates

Basically, because I want to say that you can (at least) locally transform your manifold as a local system of explicit (express coordinates in the euclidean space $\mathbb{R}^n$.

CONSEQUENCES: of the classical dynamical system

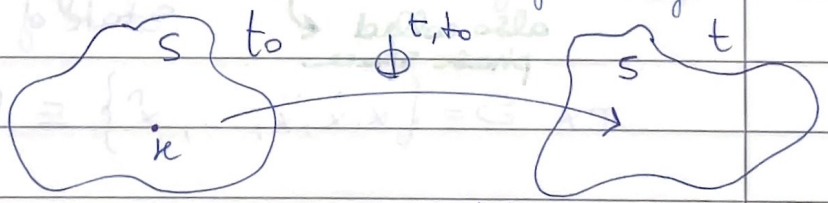
* by 2) $\Rightarrow$ the evolution law is described by a map

i.e. $\exists\ \phi : \mathcal{T} \times \mathcal{T} \times S \longrightarrow S$

s.t. $\phi(t, t_0, x)$ with $(t, t_0, x) \in \mathcal{T} \times \mathcal{T} \times S$

describes the state of the system at time t

when x is the state of the system at time t_0



Therefore, by fixing $t, t_0 \in \mathcal{T}$ $\exists\ \phi^{t,t_0} : S \longrightarrow S$

s.t. $\phi^{t,t_0}(x) := \phi(t, t_0, x)$

* and we have the following properties by 1) and 2)

i) $\phi \in C^\infty$ in $(\mathcal{T}, \mathcal{T}, S) \longmapsto S$

ii) $\phi^{t_0,t_0} = Id$, $\forall\ t_0 \in \mathcal{T}$

iii) $\phi^{t,s} \circ \phi^{s,t_0} = \phi^{t,t_0}$, $\forall\ t, s, t_0 \in \mathcal{T}$

iv) $\exists$ the inverse $(\phi^{t,t_0})^{-1} = \phi^{t_0,t}$, $\forall\ t, t_0 \in \mathcal{T}$

v) $\phi^{t,t_0} : S \longrightarrow S$ is a diffeomorfism $\forall\ t, t_0 \in \mathcal{T}$

Rmk ii) + iii) $\Rightarrow$ iv) with (t, t_0, t) and (t_0, t, t_0)

Rmk i) + iv) $\Rightarrow$ v) (i.e. diffeomorfism)

* by 3) invariance by temporal translation, i.e.

$$\circledast \quad \phi^{t,t_0} = \phi^{t+s,\,t_0+s} \quad , \forall s \in \mathcal{T}, \forall t, t_0 \in \mathcal{T}$$



Therefore, that means that I can reduce the parameters because it does matter when I am starting looking at and not how much time did pass. In other words, I can always take my initial time (when I am starting looking at) as the 0 time because it has no important what's happening previously. Therefore $\circledast$ (since it is $\forall s$) $\hookrightarrow$ then $s = 0$

$$\phi^{t-t_0,0} = \phi^{t,t_0} \quad , \forall\ t, t_0 \in \mathcal{T}$$

$\Rightarrow$ the evolution law is still described by the map

$$\phi^t := \phi^{t,0} : S \longrightarrow S, \quad t \in \mathcal{T}$$

and all the properties (i)-(v) still hold true and can be rewritten as

i') $(t,x) \longmapsto \phi^t(x) \in C^\infty$ in $\mathcal{T} \times S \longrightarrow S$

ii') $\phi^0 = Id$

iii') $\phi^t \circ \phi^s = \phi^{t+s}$, $\forall t, s \in \mathcal{T}$

iv') $\exists$ the inverse $(\phi^t)^{-1} = \phi^{-t}$, $\forall t \in \mathcal{T}$

v') $\phi^t : S \longrightarrow S$ is a diff. $\forall t \in \mathcal{T}$.

and we end up with the following definitions

DEF: The MOTION for $x \in S$ is the function $t \longmapsto \phi^t(x)$

$$\mathcal{T} \longrightarrow S$$

the ORBIT for $x \in S$ is the image $\gamma(x) := \{\phi^t(x) : t \in \mathcal{T}\}$

$$\cap$$
$$S$$

Rmk To conclude: 2) gives either the deterministic aspect of the evolution law (EL) and the reversibility.

When the (EL) is NOT reversible: that means that I cannot know the past even if from the past I can go to the future.